

Tutorials for the course BIE-MA1

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Tutorial n. 4

Continuity and derivative of a real function

Continuity of a real function; derivative and differentiability of a real function; tangent line to the graph of a function at a point; L'Hopital's rule.

Notation

sgn.....the function signum, $\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0 & x = 0 \\ -1 & x < 0 \end{cases}$

f' derivative of the function f

$f^{(n)}$ n -th derivative of the function f , $n \in \mathbb{N}$

Exercise 4.1: Plot the graph of the function $f(x) = \lfloor x \rfloor$. Investigate whether the function f is continuous, or continuous from one side (left/right)?

Solution. By definition,

$$\lfloor x \rfloor \in \mathbb{Z} : \lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$$

Therefore, for $n \in \mathbb{Z}$, $f\left(\frac{n}{2}\right) = n$ and for $x \in \left(\frac{n-1}{2}, \frac{n}{2}\right)$, $f(x) = n - 1$. From this, we can easily deduce the plot as given in picture 4.1. Since constant functions are continuous,

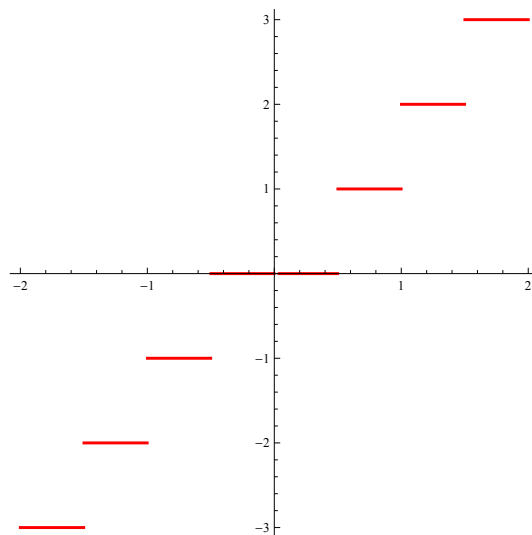


Figure 4.1: Left: Plot of $f(x) = \lfloor 2x \rfloor$.

the function f is continuous on every interval of the type $\left(\frac{n}{2}, \frac{n+1}{2}\right)$, $n \in \mathbb{Z}$. From the plot, we can infer that the function is not continuous at $x = \frac{n}{2}$ (we see what it is called discontinuity jumps at $x = \frac{n}{2}$). Let us check it formally, by using the definition of continuous function.

Recall: A function f is **continuous** at $x = x_0$ if

$$\lim_{x \rightarrow x_0} f(x) = f(x_0).$$

By studying the limit of the function at $x = \frac{n}{2}$ we see that

$$\lim_{x \rightarrow \frac{n}{2}^-} f(x) = \lim_{x \rightarrow \frac{n}{2}^-} \lfloor 2x \rfloor = n - 1 \neq f\left(\frac{n}{2}\right)$$

and

$$\lim_{x \rightarrow \frac{n}{2}^+} f(x) = \lim_{x \rightarrow \frac{n}{2}^+} \lfloor 2x \rfloor = n = f\left(\frac{n}{2}\right).$$

Therefore, the function is right-continuous at $x = \frac{n}{2}$ (but not continuous). We conclude that the function is continuous on every interval of the type $\left[\frac{n}{2}, \frac{n+1}{2}\right)$, $n \in \mathbb{Z}$.

Exercise 4.2: Sketch the graph of the function $f(x) = \operatorname{sgn}(\sin x)$. Investigate whether the function f is continuous, or continuous from one side (left/right).

Exercise 4.3: Find the derivatives of the following functions, determine the domains of the functions and of the derivatives.

$$a) f(x) = x + \sqrt{x} + \sqrt[3]{x}, \quad b) f(x) = \frac{1}{x} + \frac{2}{x^2}, \quad c) (5 + 2x)^{10}(3 - 4x)^{20}.$$

Recall: **derivative of sum, difference, product and quotient** of functions. Let the functions f and g be both differentiable at $x = a$, then

$$(f \pm g)'(a) = f'(a) \pm g'(a), \quad (f \cdot g)'(a) = f'(a)g(a) + f(a)g'(a).$$

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{(g(a))^2}, \quad \text{if } g(a) \neq 0.$$

Recall: **derivative of composition** of functions. Let the function g be differentiable at $x = a$ and the function f be differentiable at the point $g(a)$, then

$$(f \circ g)'(a) = f'(g(a))g'(a).$$

Solution. a) $D(f) = [0, +\infty)$. To find the derivative of f , we use the formula for the derivative of the sum of functions and the formula for the derivative of x^α : for $x > 0$

$$(x^\alpha)' = \alpha x^{\alpha-1}, \quad \alpha \in \mathbb{R}.$$

Notice that $\sqrt{x} = x^{\frac{1}{2}}$ and $\sqrt[3]{x} = x^{\frac{1}{3}}$. For $\alpha = 1$ we have $x' = 1$. For $\alpha = \frac{1}{2}$, we have

$$(\sqrt{x})' = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2\sqrt{x}}.$$

For $\alpha = \frac{1}{3}$ we have

$$(\sqrt[3]{x})' = \frac{1}{3}x^{\frac{1}{3}-1} = \frac{1}{3}x^{-\frac{2}{3}} = \frac{1}{3\sqrt[3]{x^2}}.$$

Thus, $D(f') = (0, +\infty)$ and

$$f'(x) = x' + (\sqrt{x})' + (\sqrt[3]{x})' = 1 + \frac{1}{2\sqrt{x}} + \frac{2}{3\sqrt[3]{x^2}}.$$

b) $f(x) = \frac{1}{x} + \frac{2}{x^2}$. $D(f) = \{x \in \mathbb{R} \mid x \neq 0\}$. We could use again the rule for the derivative of x^α , since we can write f also as $f(x) = x^{-1} + 2x^{-2}$. Or, we can use the formula for the derivative of the quotient: for $x \in D(f)$

$$\left(\frac{1}{x}\right)' = \frac{1' \cdot x - 1 \cdot x'}{x^2} = \frac{0 - 1}{x^2} = -\frac{1}{x^2}$$

and

$$\left(\frac{2}{x^2}\right)' = \frac{2' \cdot x^2 - 2 \cdot (x^2)'}{x^4} = -\frac{4x}{x^4} = -\frac{4}{x^3}.$$

Thus, $D(f') = D(f)$ and $f'(x) = -\frac{1}{x^2} - \frac{4}{x^3}$.

c) $D(f) = \mathbb{R}$. Here we can use the formula for the derivative of a product:

$$\left((5+2x)^{10}(3-4x)^{20}\right)' = \left((5+2x)^{10}\right)'(3-4x)^{20} + (5+2x)^{10}\left((3-4x)^{20}\right)'.$$

Let us focus on the computation of $\left((5+2x)^{10}\right)'$. Here we have the derivative of a composition of functions. In fact, let

$$g(x) = 5 + 2x \quad \text{and} \quad h(x) = x^{10},$$

then

$$h(g(x)) = (5 + 2x)^{10}.$$

Thus, we can apply the formula for the derivative of the composition of functions. We have

$$\left((5+2x)^{10}\right)' = h'(5+2x) \cdot (5+2x)' = 10(5+2x)^9 \cdot 2 = 20(5+2x)^9,$$

remember that $h'(x) = (x^{10})' = 10x^9$. Similarly

$$\left((3-4x)^{20}\right)' = 20(3-4x)^{19} \cdot (-4) = -80(3-4x)^{19}.$$

Therefore

$$\left((5+2x)^{10}(3-4x)^{20}\right)' = 20(5+2x)^9(3-4x)^{20} - 80(5+2x)^{10}(3-4x)^{19}.$$

Exercise 4.4: Find the derivatives of the following functions and the domains of the functions and the derivatives.

$$a) f(x) = e^{-x^2}, \quad b) f(x) = x^x, \quad c) f(x) = x^2 + 2^x.$$

Exercise 4.5: Find the derivatives of the following functions and the domains of the functions and the derivatives.

$$a) f(x) = e^{e^x}, \quad b) f(x) = 3^{x^2}.$$

$$a) f'(x) = e^{e^x} e^x, \quad b) 2 \ln(3) 3^{x^2} x$$

Exercise 4.6: Find the derivatives of the following functions and the domains of the functions and the derivatives.

$$a) f(x) = \ln(\sin x), \quad b) f(x) = \ln(\ln(\sin x)), \quad c) \operatorname{arctg} x^3, \quad d) \arcsin \frac{1}{x}.$$

$$c) f'(x) = \frac{3x^2}{1+x^6}, \quad D_f = D_{f'} = \mathbb{R}$$

$$d) f'(x) = -\frac{1}{x^2} \frac{|x|}{\sqrt{x^2-1}}, \quad D_f = (-\infty, -1] \cup [1, \infty), \quad D_{f'} = (-\infty, -1) \cup (1, \infty).$$

Exercise 4.7: Consider the function $f(x) = \sin(\ln x)$. Find $f'(1)$.

Solution. First we need to find the derivative of the function and then the value of the derivative at the point 1.

$$f'(x) = \sin'(\ln x)(\ln)'(x) = \cos(\ln x) \frac{1}{x}$$

where we used the formulas

$$\sin'(x) = \cos x \quad \text{and} \quad \ln'(x) = \frac{1}{x}.$$

Thus, we have $f'(1) = \cos(0) = 1$.

Exercise 4.8: Prove that:

$$|x|' = \begin{cases} \operatorname{sgn} x & \text{for } x \neq 0 \\ \text{does not exist} & \text{for } x = 0. \end{cases}$$

Exercise 4.9: Find the points, where the tangent line to the graph of the function

$$f(x) = \frac{1}{4}x^{5/3} + \frac{1}{3}x - 3\sqrt[3]{x}$$

is parallel to either of the axis x and y .

Exercise 4.10: Calculate the derivatives of the order 1., 2., and 3 of the following functions $f(x)$. Find the explicit formula for $f^{(n)}(x)$, for general $n \in \mathbb{N}$:

$$a) f(x) = e^x, \quad b) x^3, \quad c) x^\alpha, \alpha \in \mathbb{N}_0, \quad d) x^\alpha, \alpha \notin \mathbb{N}_0, \quad e) f(x) = \sin x, \quad f) f(x) = \cos x.$$

Exercise 4.11: Is the function f defined by the following equality

$$f(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x < 1 \\ -x^2 + 4x - 2, & 1 \leq x < 3, \\ 4 - x, & x \geq 3, \end{cases}$$

continuous on its domain?

Yes. **Solution.** $D(f) = \mathbb{R}$. The function is continuous on $(-\infty, 0) \cup (0, 1) \cup (1, 3) \cup (3, +\infty)$, since polynomial functions are continuous. We need to study the one-sided limits at the points 0, 1 and 3 to check the continuity at these points.

$$\begin{aligned}\lim_{x \rightarrow 0^-} f(x) &= \lim_{x \rightarrow 0^-} 0 = \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} x = 0 = f(0). \\ \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} x = \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} -x^2 + 4x - 2 = 1 = f(1). \\ \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} -x^2 + 4x - 2 = \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^+} 4 - x = 1 = f(3).\end{aligned}$$

The one-sided limits (i.e. left and right sided limits) are equal to the values of the function at the points 0, 1 and 3. Therefore the function is continuous on its whole domain.

Exercise 4.12: For what value of the real parameter $a \in \mathbb{R}$ is the function

$$f(x) = \begin{cases} x + 1, & x \leq 1, \\ 3 - ax^2, & x > 1 \end{cases}$$

continuous on its whole domain?

The function is continuous for $a = 1$.

Exercise 4.13: For which value of the parameters $a, b \in \mathbb{R}$ is the function

$$f(x) = \begin{cases} \frac{\tan(x)}{x}, & -\frac{\pi}{2} < x < 0 \\ a, & x = 0 \\ b(1+x)^{\frac{2}{x}}, & 0 < x < \frac{\pi}{2} \end{cases}$$

continuous on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$?

Solution. On the intervals $(-\frac{\pi}{2}, 0)$ and $(0, \frac{\pi}{2})$ the function is continuous since it equals product and quotient of elementary functions that are known to be continuous. We need to study the continuity at the point 0. We have

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{\tan(x)}{x} = \lim_{x \rightarrow 0^-} \frac{\sin x}{x} \frac{1}{\cos x} = 1$$

and

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} b(1+x)^{\frac{2}{x}} = b \lim_{x \rightarrow 0^+} \left[(1+x)^{\frac{1}{x}} \right]^2 = b \lim_{z \rightarrow +\infty} \left[\left(1 + \frac{1}{z} \right)^z \right]^2 = be^2.$$

In the last limit we used the substitution $z = \frac{1}{x} \rightarrow +\infty$ for $x \rightarrow 0^+$ and then the scheme

$$\lim_{z \rightarrow +\infty} \left(1 + \frac{1}{z} \right)^z = e.$$

The function is continuous at the point 0 if both sided limits at $x = 0$ equal the value $f(0)$, thus if

$$be^2 = 1 = a$$

that it is satisfied for $b = \frac{1}{e^2}$ and $a = 1$. For these values of the parameters the function is continuous on the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$.

Exercise 4.14: The function $f(x) = \frac{x^2 - 1}{x^3 - 1}$ is not defined at the point $x = 1$. Can the function be extended onto point $x = 1$ in the continuous way (the extension will be continuous)? If yes, find such continuous extension of f .

Solution.

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x^3 - 1} = \lim_{x \rightarrow 1} \frac{(x - 1)(x + 1)}{(x - 1)(x^2 + x + 1)} = \lim_{x \rightarrow 1} \frac{(x + 1)}{(x^2 + x + 1)} = \frac{2}{3}.$$

Since the limit at $x = 1$ exists and it has a finite value, the function can be extended by continuity onto the point $x = 1$. The extended function is given by

$$\tilde{f}(x) = \begin{cases} f(x), & x \neq 1 \\ \frac{2}{3}, & x = 1 \end{cases}$$

Exercise 4.15: The function $f(x) = \arctg \frac{1}{x}$ is not defined at the point $x = 0$. Can the function be extended onto point $x = 0$ in the continuous way (the extension will be continuous)? If yes, find such continuous extension of f .

No. $\lim_{x \rightarrow 0+} f(x) = \frac{\pi}{2}$ and $\lim_{x \rightarrow 0-} f(x) = -\frac{\pi}{2}$.

Exercise 4.16: For which values of the real parameters $a \in \mathbb{R}$ can the function

$$f(x) = \frac{\ln(1 + ax^2)}{x}$$

be extended onto $x = 0$ by continuity? In the cases when it is possible, find the continuous extension of f .

Solution: f can be extended by continuity $\forall a \in \mathbb{R}$. The continuous extension is given by

$$\tilde{f}(x) = \begin{cases} f(x), & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Exercise 4.17: For which values of the real parameters $a, b \in \mathbb{R}$ can the function

$$f(x) = \begin{cases} ax - 1, & x > 1 \\ b \frac{e^{x-1} - 1}{(x-1)^2} & x < 1 \end{cases}$$

be extended onto $x = 1$ by continuity? In the cases when it is possible, find the continuous extension of f .

Solution: f can be extended for $a = 1, b = 0$. The continuous extension is given by

$$f(x) = \begin{cases} x - 1, & x > 1 \\ 0, & x \leq 1 \end{cases}$$

Exercise 4.18: Is the function

$$f(x) = \begin{cases} -1 + \cos 3x, & x \geq 0 \\ e^x - 1 & x < 0 \end{cases}$$

continuous and differentiable at $x = 0$?

Recall: A function is differentiable at a point $a \in D(f)$ if

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists and it has a finite value. The above limit is equivalent to

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Recall: If a function f is differentiable at $x = a$ then it is continuous (by the thm. on the relationship between differentiability and continuity). The converse is not true, i.e. a function can be continuous at $x = a$ but not differentiable at $x = a$.

Solution. On the intervals $(-\infty, 0)$ and $(0, +\infty)$ the function is continuous differentiable since it equals difference of elementary functions (or their composition) that are known to be continuous and differentiable. We need to study the continuity and differentiability at the point 0.

Let us start by studying the differentiability at $x = 0$:

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} &= \lim_{x \rightarrow 0^+} \frac{-1 + \cos(3h)}{h} = \lim_{h \rightarrow 0^+} \frac{-1 + \cos^2(3h)}{h(1 + \cos(3h))} \\ &= \lim_{h \rightarrow 0^+} \frac{\sin^2(3h)}{h^2} \frac{h}{(1 + \cos(3h))} = 0. \end{aligned}$$

$$\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} = \lim_{x \rightarrow 0^-} \frac{e^h - 1}{h} = 1.$$

Since the above left and right-side limits are not equal, the function is not differentiable at $x = 0$, therefore it is not differentiable on its whole domain. We still have to study the continuity at $x = 0$:

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} e^x - 1 = 0 = f(0).$$

Thus the function is continuous at $x = 0$ and therefore it is continuous on its whole domain.

Exercise 4.19: For which values of the real parameters $a, b \in \mathbb{R}$ is the function

$$f(x) = \begin{cases} ax + 1, & x \geq 1 \\ 1 - bx^2 & x < 1 \end{cases}$$

continuous and differentiable on its domain?

Exercise 4.20: For which values of the real parameters $a, b \in \mathbb{R}$ is the function

$$f(x) = \begin{cases} a + \sqrt{x-2}, & x \geq 2 \\ 1 - b(x-2)^2 & x < 2 \end{cases}$$

continuous and differentiable on its domain?

Exercise 4.21: For which values of the real parameters $a, b \in \mathbb{R}$ is the function

$$f(x) = \begin{cases} b, & x \geq 0 \\ \ln|x|^a + 1 & x < 0 \end{cases}$$

continuous and differentiable on its domain?

Exercise 4.22: For which values of the real parameters $a, b \in \mathbb{R}$ is the function

$$f(x) = \begin{cases} bx, & x > 0 \\ 0, & x = 0 \\ e^{|x|^a} - 1 & x < 0 \end{cases}$$

continuous and differentiable on its domain?

Solution. The function f is differentiable, and therefore also continuous, on the intervals $(-\infty, 0)$ and $(0, +\infty)$, since on these intervals f equals product or difference of elementary functions (or their composition) which are known to be differentiable. We have to study the continuity and differentiability at $x = 0$. For the continuity at $x = 0$:

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} bx = 0 = f(0), \quad \forall b \in \mathbb{R}$$

and

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} e^{(-x)^a} - 1 = \begin{cases} 0 & a > 0 \\ e - 1, & a = 0 \\ +\infty & a < 0 \end{cases}.$$

Thus the function is not continuous, and therefore not differentiable, at $x = 0$ if $a \leq 0$ and it is continuous at $x = 0$ for every value of b if $a > 0$. We study the differentiability only if the function is continuous, i.e. for $a > 0, b \in \mathbb{R}$. We find

$$\lim_{x \rightarrow 0^+} \frac{bx}{x} = b$$

and

$$\begin{aligned} \lim_{x \rightarrow 0^-} \frac{e^{(-x)^a} - 1}{x} &= \lim_{x \rightarrow 0^-} \frac{(-x)^a}{x} \cdot \frac{e^{(-x)^a} - 1}{(-x)^a} = - \lim_{x \rightarrow 0^-} \frac{(-x)^a}{-x} \\ &= - \lim_{x \rightarrow 0^-} (-x)^{a-1} = \begin{cases} 0, & a > 1 \\ -1, & a = 1 \\ -\infty & 0 < a < 1 \end{cases} \end{aligned}$$

Thus, the function is differentiable at $x = 0$, and therefore on its whole domain, for $a > 1$ and $b = 0$ or for $a = 1$ and $b = -1$.

Exercise 4.23: For which values of the real parameters $a, b \in \mathbb{R}$ is the function

$$f(x) = \begin{cases} \sin 2x, & x < 0 \\ b, & x = 0 \\ e^{x^a} - 1 & x > 0 \end{cases}$$

continuous and differentiable on its domain?

Exercise 4.24: Find the derivatives of the following functions

$$a) f(x) = \operatorname{arctg} e^x, \quad b) f(x) = x \sin(x) \cos(x), \quad c) f(x) = \ln |x|, \quad d) f(x) = \frac{1}{\arcsin(x)}.$$

$$a) f'(x) = \frac{e^x}{1+e^{2x}}, \quad b) f'(x) = \frac{1}{2} \sin(2x) + x \cos(2x), \quad c) f'(x) = \frac{1}{x}, \quad d) f'(x) = -\frac{1}{(\arcsin(x))^2} \frac{1}{\sqrt{1-x^2}}.$$

Exercise 4.25: For the function f defined below, find the points $a \in D(f)$ such that the angle between the X -axis and the tangent line to the function at the point a is α :

$$\begin{aligned} a) f(x) &= x^3, \quad \alpha = \frac{\pi}{3}, \\ b) f(x) &= (x^2 - x)^{1/3}, \quad \alpha = \frac{\pi}{2}, \\ c) f(x) &= -\operatorname{arctg}(x), \quad \alpha = \frac{7\pi}{4} \end{aligned}$$

$$a) a = \pm 3^{-1/4}, \quad b) a \in \{0, 1\}, \quad c) a = 0.$$

Exercise 4.26: Find the derivatives of the following functions (a, b, c, d are positive real parameters)

$$\begin{aligned} a) f(x) &= \frac{a}{x} + \frac{x}{a} + \frac{b^2}{x^2} + \frac{x^2}{b^2}, \quad b) f(x) = \left(\frac{1}{2} - x\right)^2, \quad c) f(x) = \sqrt{x} (x^3 - \sqrt{x} + 1), \\ d) f(x) &= \frac{ax+b}{cx+d}, \quad e) f(x) = (\sqrt{x} + 1) \left(\frac{1}{\sqrt{x}} - 1\right), \quad f) f(x) = (x^2 - 3x + 3)(x^2 + 2x - 1). \end{aligned}$$

$$\begin{aligned} a) f'(x) &= -\frac{a}{x^2} + \frac{1}{a} - 2\frac{b^2}{x^3} + 2\frac{x}{b^2}, \quad b) f'(x) = 2x - 1, \quad c) f'(x) = \frac{1}{2\sqrt{x}} (7x^3 - 2\sqrt{x} + 1), \quad d) f'(x) = \frac{ad-bc}{(cx+d)^2}, \quad e) \\ f'(x) &= -\frac{1+x}{2x^{3/2}}, \quad f) f'(x) = 4x^3 - 3x^2 - 8x + 9. \end{aligned}$$

Exercise 4.27: For the function $f(x) = 3x - 2\sqrt{x}$ find the values $f(1)$, $f'(1)$, $f(4)$ and $f'(4)$. Find the formulas for $f(a^2)$ and $f'(a^2)$ ($a \in \mathbb{R}$).

$$f(1) = 1, \quad f'(1) = 2, \quad f(4) = 8, \quad f'(4) = \frac{5}{2}, \quad f(a^2) = 3a^2 - 2|a|, \quad f'(a^2) = 3 - \frac{1}{|a|}.$$

Exercise 4.28: For the function $f(x) = \frac{x^2-5x-1}{x^3}$ calculate the values $f(-1)$, $f'(-1)$ and $f'(2)$. Find an explicit formula for $f'\left(\frac{1}{a}\right)$, where $a \neq 0$.

$$f(-1) = -5, \quad f'(-1) = -8, \quad f'(2) = \frac{19}{16}, \quad f\left(\frac{1}{a}\right) = a^2(3a^2 + 10a - 1)$$

Exercise 4.29: Find the derivatives of the following functions

$$\begin{aligned} a) f(x) &= \frac{x}{1 - \cos(x)}, \quad b) f(x) = x \sin x + \cos x, \quad c) f(x) = \sin(\sin(x)), \quad d) f(x) = \sqrt{\operatorname{tg} \frac{x}{2}}, \\ e) f(x) &= \sin \sqrt{1+x^2}, \quad f) f(x) = \arcsin \frac{2}{x}, \quad g) f(x) = \frac{x}{1+x^2} - \operatorname{arctg} x. \end{aligned}$$

$$\begin{aligned} a) f'(x) &= \frac{1 - \cos(x) - x \sin(x)}{(1 - \cos(x))^2}, \quad b) f'(x) = x \cos(x), \quad c) f'(x) = \cos(\sin(x)) \cdot \cos(x), \quad d) f'(x) = \frac{1}{4} \frac{1}{\left(\sin \frac{x}{2}\right)^{1/2} \left(\cos \frac{x}{2}\right)^{3/2}}, \\ e) f'(x) &= \frac{x \cos \sqrt{1+x^2}}{\sqrt{1+x^2}}, \quad f) f'(x) = -\frac{2}{|x|\sqrt{x^2-4}}, \quad g) f'(x) = -2 \left(\frac{x}{1+x^2}\right)^2. \end{aligned}$$

Exercise 4.30: Find the derivatives of the following functions (a, b are real parameters)

$$\begin{aligned} a) f(x) &= 2^x, & b) f(x) &= x \cdot 10^x, & c) f(x) &= 2^{\frac{x}{\ln(x)}}, & d) f(x) &= \frac{1+e^x}{1-e^x}, \\ e) f(x) &= ae^{-b^2x^2}, & f) f(x) &= a^x x^a \text{ (zde } a > 0), & g) f(x) &= 2^{3^x}, & h) f(x) &= e^{\sqrt{x+1}}. \end{aligned}$$

$$\begin{aligned} a) f'(x) &= 2^x \ln(2), & b) f'(x) &= (1+x \ln(10)) \cdot 10^x, & c) f'(x) &= \frac{\ln(2)}{\ln^2(x)} (\ln(x)-1) 2^{\frac{x}{\ln(x)}}, & d) f'(x) &= \frac{2e^x}{(e^x-1)^2}, \\ e) f'(x) &= -2ab^2xe^{-b^2x^2}, & f) f'(x) &= (x \ln(a) + a) a^x x^{a-1}, & g) f'(x) &= 2^{3^x} \cdot \ln(2) \cdot 3^x \cdot \ln(3), & h) f'(x) &= e^{\sqrt{x+1}} \frac{1}{2\sqrt{x+1}}. \end{aligned}$$

Exercise 4.31: By using l'Hospital's rule, calculate the following limits

$$a) \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}, \quad b) \lim_{x \rightarrow 1} \sin(x-1) \operatorname{tg} \frac{\pi x}{2}, \quad c) \lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$$

Solution.

$$\begin{aligned} a) \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x} &= \left[\frac{0}{0} \right] \stackrel{\text{H.}}{=} \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - 2}{1 - \cos x} = \left[\frac{0}{0} \right] \\ &\stackrel{\text{H.}}{=} \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin x} = \left[\frac{0}{0} \right] \stackrel{\text{H.}}{=} \lim_{x \rightarrow 0} \frac{e^x + e^{-x}}{\cos x} = 2 \\ b) \lim_{x \rightarrow 1} \sin(x-1) \tan \frac{\pi x}{2} &= [0 \cdot \infty] = \lim_{x \rightarrow 1} \frac{\sin(x-1) \sin \frac{\pi x}{2}}{\cos \frac{\pi x}{2}} = \left[\frac{0}{0} \right] \\ &= \lim_{x \rightarrow 1} \frac{\sin(x-1)}{\cos \frac{\pi x}{2}} \stackrel{\text{H.}}{=} \lim_{x \rightarrow 1} \frac{\cos(x-1)}{-\frac{\pi}{2} \sin \frac{\pi x}{2}} = -\frac{2}{\pi} \\ c) \lim_{x \rightarrow 1} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right) &= \lim_{x \rightarrow 1} \frac{x-1-\ln x}{\ln x(x-1)} = \left[\frac{0}{0} \right] \stackrel{\text{H.}}{=} \lim_{x \rightarrow 1} \frac{1-\frac{1}{x}}{\frac{1}{x}(x-1)+\ln x} = \left[\frac{0}{0} \right] \\ &= \lim_{x \rightarrow 1} \frac{\frac{x-1}{x}}{\frac{x-1+x \ln x}{x}} = \lim_{x \rightarrow 1} \frac{x-1}{x-1+x \ln x} = \left[\frac{0}{0} \right] \stackrel{\text{H.}}{=} \lim_{x \rightarrow 1} \frac{1}{2+\ln x} = \frac{1}{2}. \end{aligned}$$

Exercise 4.32: By using l'Hospital's rule, calculate the limits in the following exercises of Tutorial 4: 4.13, 4.14, 4.15, 4.19, 4.20, 4.21, 4.22, 4.24, 4.26, 4.27, 4.29, 4.31, 4.32, 4.33, 4.34.